

Data Science for Economists

Lecture 1: Introduction

Jiaxi Li

University of North Carolina | ECON 370

Table of contents

1. Prologue
2. Syllabus highlights
3. Getting started
4. R for data science
5. Data visualization with ggplot2

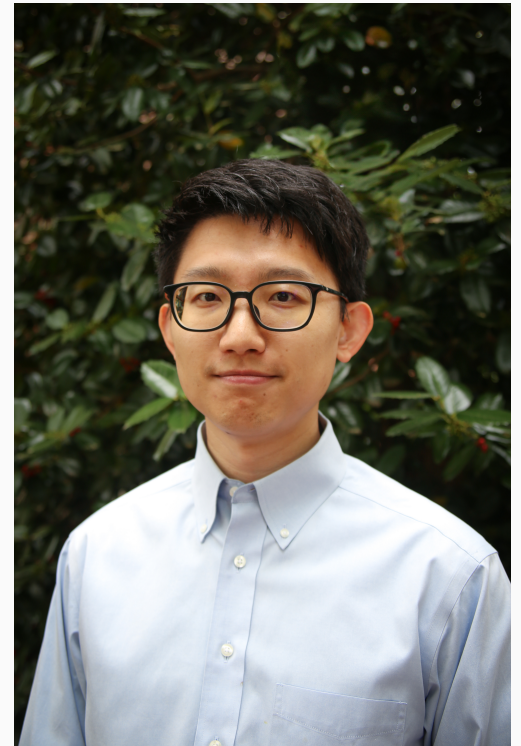
* Slides adapted from Drew Van Kuiken and Alex Marsh's ECON 370 and originally based on materials by Grant McDermott's EC 607 at University of Oregon.

Prologue

Introductions

Me

- Jiaxi Li
- lijiaxi@unc.edu (I will email you back as fast as possible.)
- Office Hours: Th 2:15-3:15PM (Does this work for people?)
- Office: Gardner Hall 5th floor library (during Office Hours)
- I'm from China (and North Carolina)
- Now: Fifth Year Economics PhD Student (Financial Econometrics)
- Undergraduate: Here at UNC! (Applied Math and Economics, minor in Statistics)
- Foreign Exchange: One semester in Tübingen, Germany
- Masters: UChicago



Introductions

Your Turn

- Full Name (Preferred Name): [Your Name]
- Hometown / Home Country: [City, State or Country]
- Major / Academic Year: [Your Major], [e.g., Junior / 3rd Year Undergraduate]
- Relevant Coursework in Economics: [List economics courses taken]
- Relevant Coursework in Statistics: [List statistics courses taken]
- Programming Experience: [Briefly describe tools/languages used and in what context — e.g., coursework, projects, research]

Note: I have collected these information in the survey! If you have not yet done so, please do it [here](#).

First Dataset!

We just created our first dataset in this class. Now, let's take a look and do a little analysis.

Small datasets: visual inspection or intuition (not quite accurate) or statistics

Large datasets: appropriate programming tools (such as R)

Syllabus Highlights

(Read the full document [here](#).)

ECON 370

Economic Applications of Data Science

Comparing Data Science, Econometrics, Economics (Theory), and Computer Science:

- *Data Science (Statistics)*: Focuses on analyzing data to learn about the world — often without a guiding theory.
- *Economics Theory*: Develops theoretical frameworks to explain human behavior and societal dynamics.
- *Econometrics*: Applies statistical methods to test economic theories or answer empirical economic questions.
- *Computer Science*: Involves designing computer programs to perform specific tasks such as building games, applications, or websites.

Econ 370 vs. Econ 400:

- *Econ 400*: emphasizes *econometrics theory and statistical analysis*
- *Econ 370*: places greater emphasis on *data cleaning and computational implementation*

Why this course?



Fill in the gaps left by traditional econometrics and methods classes.

- Practical skills and tools that will complement your other econometric courses.
- Neglected skills like how to actually find datasets in the wild and clean them.
- It also matters for your job! (sometimes)

Data science skills are largely distinct from (and complementary to) the core 'metrics curriculum familiar to economists.

- Data viz, cleaning and wrangling; programming; cloud computation; relational databases; machine learning; etc.

Grading

Component	Weight
5 × homework assignments	60%
1 × final project	30%
Participation	10%

- Number of homework assignments might change.
 - Regardless, will be equally weighted.
 - Individual weight = $60/n$.
- Homework assignments maybe done in groups of 3 or 4. However, discussing concepts and comparing approaches with peers is encouraged.
 - If done in groups all group members are still expected to program on their own.
 - If done in groups, please submit only one assignment with all names.
- I will provide you with R Markdown templates for assignments and require you to submit assignments with an R Markdown file.
 - Discuss details when the first assignment comes up.
- Final Project
 - About Predicting Stock Returns, more details later.
 - Individual project, short presentations during our final period.

Course Overview

Course Goals

1. Teach students how to competently program in R with good style,
2. Teach students how to think and approach problems from a computational perspective,
3. Teach students basic data science skills including data visualization and basic models,
4. Imbue students with a desire to learn more about econometrics and data science.

Course Overview

Learning Objectives

1. Be able to write functioning, readable, and aesthetically pleasing code in the R programming language,
2. Give raw data, be able to manipulate the data into the correct format needed for an analysis,
3. Given data and a research question, be able to create a exploratory data visualization in the right format to get at answering the question,
4. Be able to communicate results to a non-technical audience.

Lecture outline

Introduction

- Introduction: Motivation, software installation, and data visualization
- R language basics (will take some time)

Applications of Numerical Programming

Topics here are good to know but are mostly to practice programming

- Simulation
- Optimization

Data skills

- `data.table` and `dplyr`
- Merging

Lecture outline (cont.)

Basics of Data Science

- Data Visualization
- Regression
- Classification
- Machine Learning and Artificial Intelligence

Textbooks and Resources

I will be pulling from various books that are publicly available.

- *R for Data Science (2e)* by Hadley Wickham, Mine Cetinkaya-Rudel, and Garrett Golemund
- *Hands-On Programming with R* by Garrett Golemun
- *An Introduction to Statistical Learning with Applications in R* by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani

Additional Books and Resources

- *Learning R* by Richard Cotton
- *Advanced R* by Hadley Wickham
- *Swirl: Learn R, in R* -- Great resource to learn the basics of R within R
- *R Markdown Cookbook*
- UNC Library [BeginR](#) and [Open Lab GitHub page](#)

Note: UNC library also offer crash courses for other languages. [Here](#) is a old post but there should be something similar.

Other Ways to Learn Programming

- More than anything else, learning to code is done via trial and error

1. StackOverflow:

- Online message board, truly incredible number of questions and answers to comb through
- I've never posted on StackOverflow, but you might want to! If you do, make sure you follow their rules
- Usually, a couple different threads will pertain to your question - review all of them to get a sense of how to solve your issue

2. LLMs

- I use ChatGPT to help me understand error messages and find possible solutions
- I also use it to explain some function or code I am not familiar with (in other languages as well)
- ChatGPT sometimes sucks at coding. Check the AI's work!
- Make sure you understand why something works

3. Various presentations and blog posts on the internet

- e.g., [Comparison of dplyr and data.table](#)
- Avoid TowardsDataScience, websites with tons of ads

Getting started

Software installation and registration

1. Download **R**.
2. Download **RStudio**.



If you had trouble completing any of these steps, please raise your hand.

- See *Appendix A of Hands-On Programming with R*

Some OS-specific extras

I'll detail further software requirements as and when the need arises. However, to help smooth some software installation issues further down the road, please also do the following (depending on your OS):

- **Windows:** Install **Rtools**.
- **Mac:** Might need to install Xcode later. This can take awhile, so I will let you know.
- **Linux:** None (you should be good to go).

Checklist

- ☑ Do you have the most recent version of R?

```
version$version.string
```

```
## [1] "R version 4.5.1 (2025-06-13 ucrt)"
```

- ☑ Do you have the most recent version of RStudio? (The **preview version** is fine.)

```
RStudio.Version()$version
```

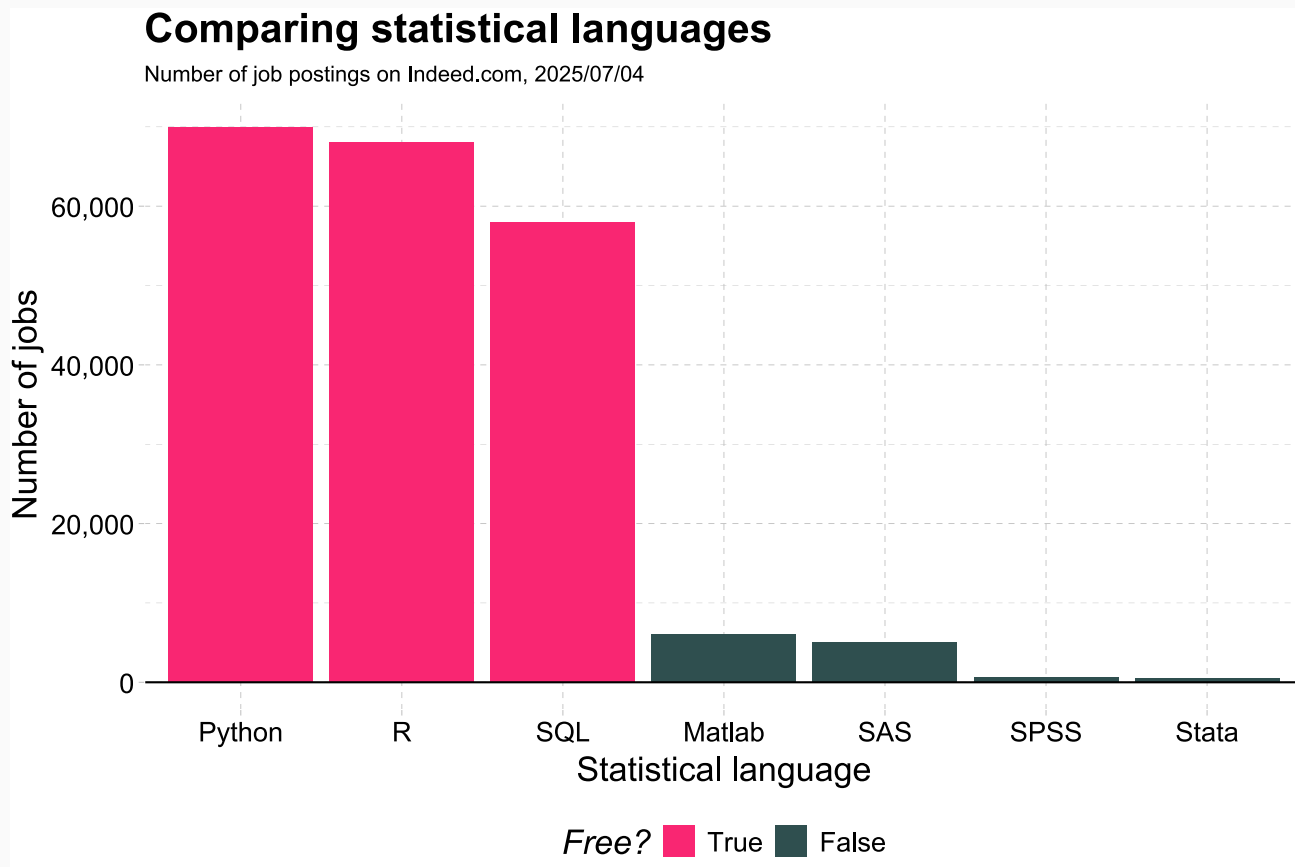
```
## Requires an interactive session but should return something like "[1] '2024.4.2..
```

- ☑ Have you updated all of your R packages?

```
update.packages(ask = FALSE, checkBuilt = TRUE)
```

R for data science

Why R and RStudio? (cont.)



Why This Course? Why R? My Experience:

- What I "learned" in college: Matlab, Mathematica, Stata, R, Excel
- Mathematica is too specialized for mathematics, Stata is more like a calculator and Excel can only do simple stuff (imo)
- What I use now: R, Python, Matlab, Julia
- Languages change over time: for now, R is probably your best bet
- Being flexible is most valuable - ultimately, you want to learn how computers think
- Computer languages are how we talk to computers—so precision matters!
- Different languages often have similar logic! Learning one can help you understand other ones.

Why R and RStudio? (cont.)

Data science positivism

- Alongside Python, R has become the *de facto* language for data science.
 - See: *The Impressive Growth of R*, *The Popularity of Data Science Software*
- Open-source (free!) with a global user-base spanning academia and industry.
 - LLMs are good at it! (rip Matlab)

Bridge to applied economics and other tools

- Already has all of the statistics and econometrics support, and is amazingly adaptable as a “glue” language to other programming languages and APIs.
- The RStudio IDE and ecosystem allow for further, seamless integration.

Pros and Cons of R

- ✓ Free.
- ✓ Great community with lots of support.
- ✓ Syntax is fairly basic for those with a math background.
- ✓ RStudio.
- ✓ Beautiful Graphics (better than Python 😊).
- ✓ Data wrangling.
- ✗ Slow. (Only slightly comparing to Julia.)
- ✗ Demanding on RAM.
- ✗ Dependency on packages.

R vs Python

Both R and Python are common data science languages.

However, Python is arguably the more important language in industry.

So why not learn Python for this class?

1. I think R is the easier of the two to get started learning.
2. There is much commonality between the two.
3. R is better for learning statistics as it is ultimately a programming language designed with statistical computing in mind.
4. R has much cooler visualizations with ggplot2.

R is a language for statistics whereas Python is a language that can do statistics.

[Here](#) is a YouTube video comparing those two languages.

Laptops in Class

We'll be using our laptops a lot in class this semester. This is the best way to learn how to code!

Playing games, watching sports, shopping online are *incredibly* distracting. *Please!!!!!!* do not do this.

If you are unable to control yourself, *please* sit in the back of the classroom

I reserve the right to take participation points off if you're making life harder for your classmates.

A Note on File Structure

It has come to my attention that many college students do not use folders

When I was in high school, I did not use folders.

Things are extremely hard to find in a large pile of files.

A Suggested File Structure

D: → R → ECON370 → hw1, final project, books, slides, etc.

For any coding project (and, e.g., within the hw1 folder) use the following subfolders (trust me on this):

- data
- raw_data
- scripts
- output
- temp

Notes on folders:

- **No spaces in folder names!!!**
- Within folders like scripts, create subfolders named _old. Don't delete old files, just move them to _old
- When creating files with multiple versions, it is useful to have filenames with the date or version number. HW1_V1.R or HW1_2025_08_19.R

Directory and Default Directory

A directory is the folder where R is currently working. R will load files from, and save files to, this location-known as the current working directory.

You can change it in RStudio by navigating to the desired folder, then selecting:

More → Set As Working Directory

After this, any files you save will appear in the newly set working directory.

To avoid having to manually navigate to the correct folder each time, you can set a default working directory-a convenient location that contains all your R files.

Go to:

Tools → Global Options → change the Default working directory → Apply

From then on, every time you open RStudio, it will start in your chosen default folder.

A Lay of the Land

R is a coding language. If you want, you can just open your computer's terminal and start typing in R.* This is like directly tell computer/people do some thing, like "1+1".

RStudio is the studio for R. (Like a studio for woodworking) You can find all helpful things here for R programming.

1. You can start typing R code into the console and make your computer do stuff.
2. You can also write script in R files called *R Script*. (*File → New File → R Script*)

This is like using the English language to write a book. You keep a copy about the things you want to say. But you still have to run things in the console (or Highlight and then Control&Enter). (Read your written speech to the computer.)

3. You can also write both word and code in R called *R Markdown*. Then you knit it into a single file. (*File → New File → R Markdown...*)

Markdown is still widely used. All slides in this class are written with R Markdown!

4. The newer version of R Markdown is Quarto. Quarto visual shows you the output look while coding. Quarto source is basically Markdown. (*File → New File → R Quarto...*)

More on R Markdown (.Rmd)

What is R Markdown?

R Markdown is a file format in RStudio that lets you combine:

- **Text** (like a Word document)
- **R code** (that you can run)
- **The results of that code** (tables, plots, analysis)

It's like a notebook where your explanations, code, and results all live together.

Why use R Markdown?

1. Reproducibility - You can re-run the document and get the exact same results later.
2. Transparency - Code and results are side-by-side, so others can see how you did your analysis.
3. Easy reporting - Turn your file into HTML, PDF, or Word documents with a single click.
4. Efficiency - Update your analysis once and automatically refresh all results.

How it works

Go to:

File → New File → R Markdown ...

Note: You can click `visual` if you think that interface is more aesthetically pleasing.

An R Markdown file has three parts:

1. YAML Header (top part in `---`): sets the title, author, date, and output format.
2. Text Sections: written in Markdown — a simple way to format text (headings, bold, lists, etc.).
3. Code Chunks: blocks of R code inside triple backticks.

Running the document

Click the `knit` button in RStudio.

`R` will run all code chunks in order and produce a finished document (HTML, PDF, or Word).

If something breaks, fix the code and knit again.

We can use this to create project report, paper, presentation, etc. (More about this in lec5)

Additional tips:

- Write as you go: Don't separate "doing the work" and "writing the report."
- Keep it clean: Comment your code so future-you understands it.

Control chunk options:

- `echo = FALSE`: hides the code, shows only results.
- `eval = FALSE`: shows the code, doesn't run it.
- `message = FALSE`, `warning = FALSE`: hides messages and warnings.

A Lay of the Land

Note: R is built based on English. Try to avoid foreign characters.

Nowadays, there are three paradigms people use to write R code:

1. Using base R
2. Using the tidyverse (a *package* with a huge set of functions for writing code)
3. Using data.table (another *package* with a huge, different set of functions for writing code)

A Lay of the Land

Note: R is built based on English. Try to avoid foreign characters.

Nowadays, there are three paradigms people use to write R code:

1. Using base R

- Section 1 is in base R
- Think of this as an introduction to coding itself. You likely won't write much base R going forward, but the principles we learn here will pop up everywhere
- Learn counting before addition!!!

2. Using the tidyverse (a *package* with a huge set of functions for writing code)

3. Using data.table (another *package* with a huge, different set of functions for writing code)

A Lay of the Land

Note: R is built based on English. Try to avoid foreign characters.

Nowadays, there are three paradigms people use to write R code:

1. Using base R

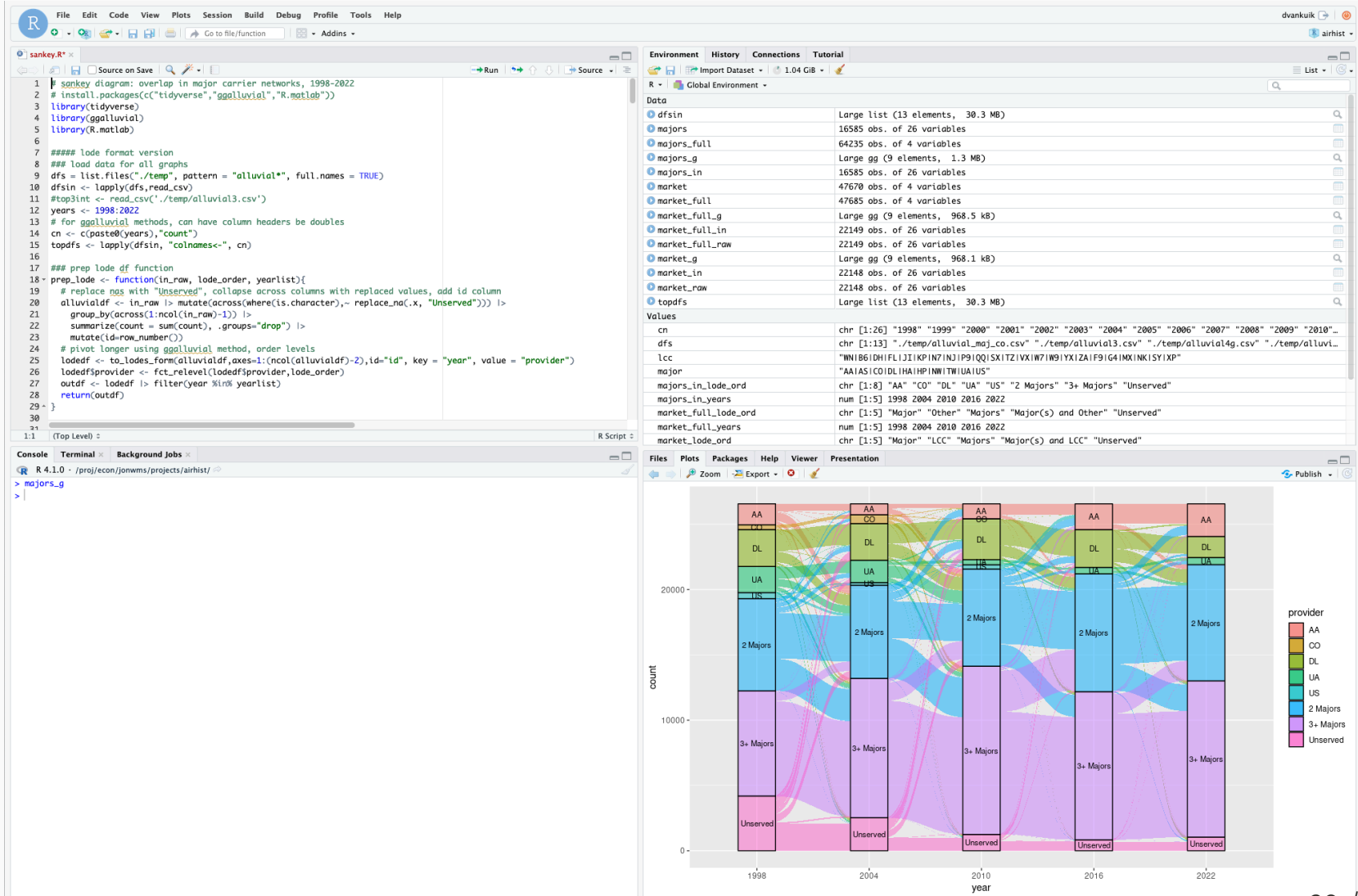
- Section 1 is in base R
- Think of this as an introduction to coding itself. You likely won't write much base R going forward, but the principles we learn here will pop up everywhere
- Learn counting before addition!!!

2. Using the tidyverse (a *package* with a huge set of functions for writing code)

3. Using data.table (another *package* with a huge, different set of functions for writing code)

- Section 2 will introduce the **tidyverse** and **data.table** packages
- Pipeline in **tidyverse** makes your code cleaner and more efficient by reducing long lines and unnecessary intermediate objects.
- I suggest you write most of your code using the **tidyverse**. It's easy to pick up and very readable
- You'll see lots of internet code using **data.table**. It's probably more powerful on net than the **tidyverse**, so it's important you're familiar with it

A Lay of the Land (Virtual)



Some R basics

1. Everything is an object.
2. Everything has a name.
3. You do things using functions.
4. Functions come pre-written in packages (i.e. "libraries"), although you can — and should — write your own functions too.
5. Use # for commenting. Things would not run if there is a # in the front. Always leave comments for future reference!!!!

Points 1. and 2. can be summarized as an **object-oriented programming** (OOP) approach.

- This may sound super abstract now, but we'll see *lots* of examples over the coming weeks that will make things clear.

Some useful short keys

- Highlight multiple lines + Control&Shift&c → Comment/Uncomment multiple lines
- Highlight multiple lines + Control&Enter → Run multiple lines of code
- Highlight multiple lines + tab → indent multiple lines
- Highlight multiple lines + Shift&tab → unindent multiple lines
- Control&z → Undo changes
- Control&Shift&z → Redo changes
- Control&f → Find and replace

Here is a more complete list of short keys.

A line of # can even divide sections of the code and act as a break line!

Warning about the Insert key:

If you accidentally press the *Insert* key, your typing may start overwriting existing code. Don't panic—just press the *Insert* key again to return to normal editing mode.

R code example (linear regression)

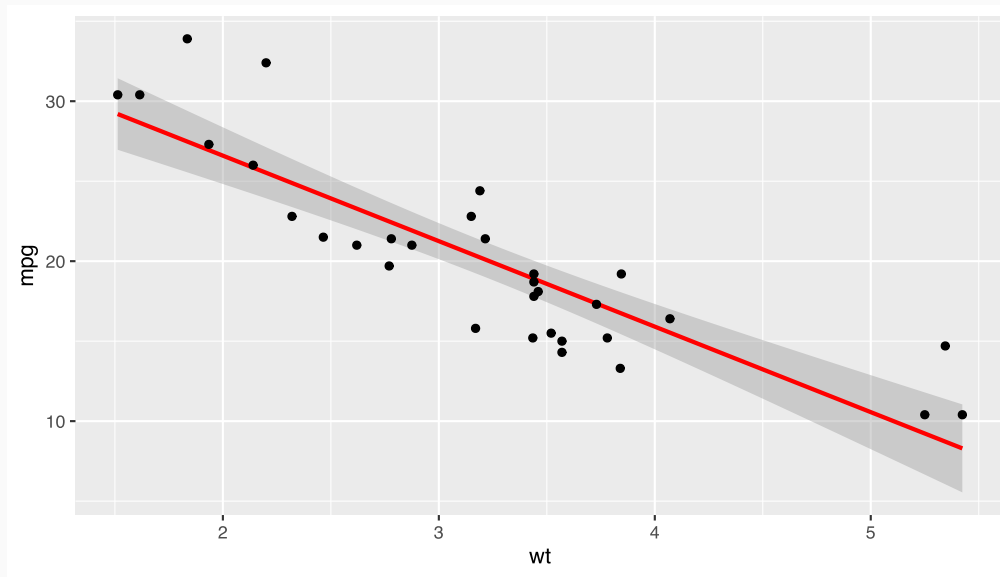
```
fit = lm(mpg ~ wt, data = mtcars)
summary(fit)

##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851     1.8776   19.858 < 2e-16 ***
## wt          -5.3445     0.5591   -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
## Multiple R-squared:  0.7528,    Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

ggplot2

```
library(ggplot2)
ggplot(data = mtcars, aes(x = wt, y = mpg)) +
  geom_smooth(method = "lm", col = "red") +
  geom_point()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



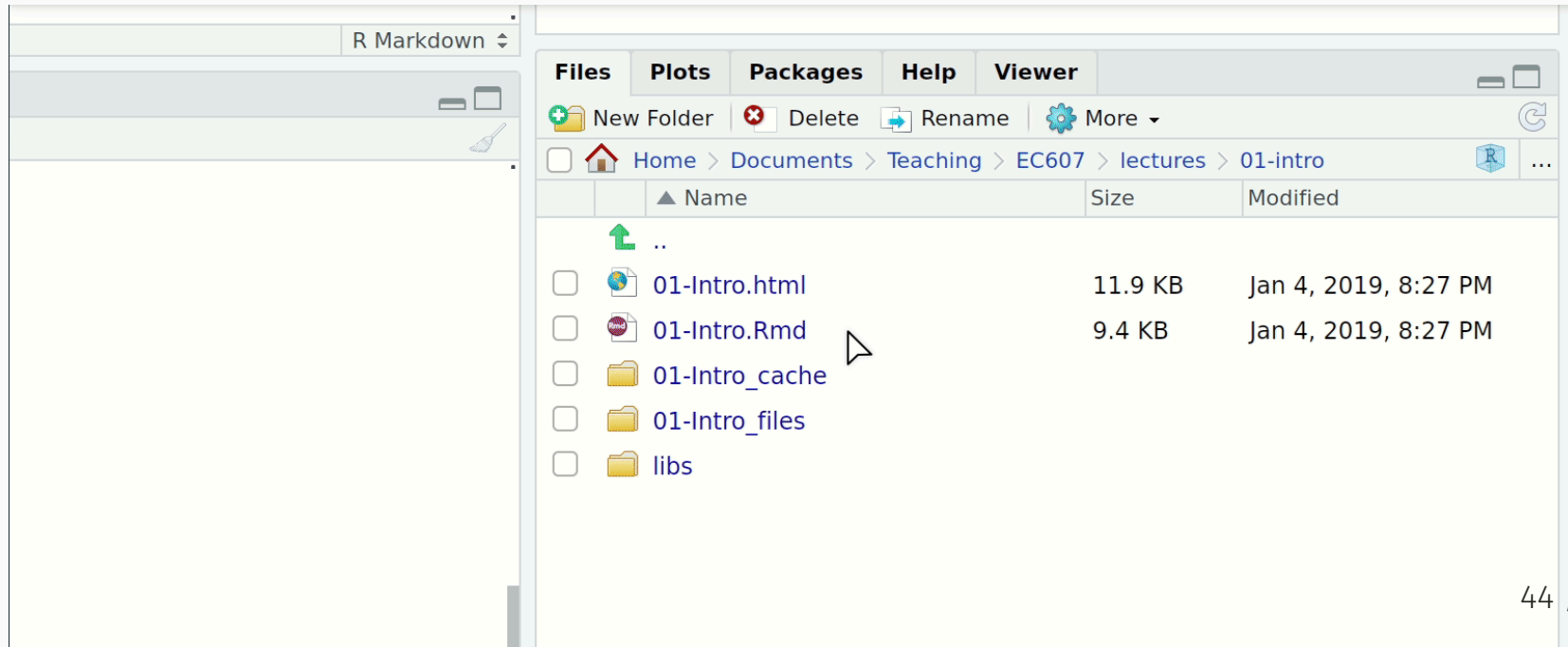
More ggplot2

Install and load

Open up your laptops. For the remainder of this first lecture, we're going to continue playing around with `ggplot2` (i.e. livecoding).

If you don't have them already, install the `ggplot2` and `gapminder` packages via either:

- **Console:** Enter `install.packages(c("ggplot2", "gapminder"), dependencies=T)`.
- **RStudio:** Click the "Packages" tab in the bottom-right window pane. Then click "Install" and search for these two packages.



Install and load (cont.)

Once the packages are installed, load them into your R session with the `library()` function.

```
library(ggplot2)
library(gapminder) ## We're just using this package for the gapminder data
```

Notice too that you don't need quotes around the package names any more. Reason: R now recognises these packages as defined objects with given names. ("Everything in R is an object and everything has a name.")

PS — A convenient way to combine the package installation and loading steps is with the **pacman package's** `p_load()` function. If you run `pacman::p_load(ggplot, gapminder)` it will first look to see whether it needs to install either package before loading them. Clever.

- We'll get to this next week, but if you want to run a function from an (installed) package without loading it, you can use the `PACKAGE::package_function()` syntax.

Brief aside: The gapminder dataset

Because we're going to be plotting the **gapminder** dataset, it is helpful to know that it contains panel data on life expectancy, population size, and GDP per capita for 142 countries since the 1950s.

```
gapminder
```

```
## # A tibble: 1,704 × 6
##   country      continent  year lifeExp      pop gdpPercap
##   <fct>        <fct>    <int>  <dbl>    <int>    <dbl>
## 1 Afghanistan Asia      1952   28.8  8425333    779.
## 2 Afghanistan Asia      1957   30.3  9240934    821.
## 3 Afghanistan Asia      1962   32.0 10267083    853.
## 4 Afghanistan Asia      1967   34.0 11537966    836.
## 5 Afghanistan Asia      1972   36.1 13079460    740.
## 6 Afghanistan Asia      1977   38.4 14880372    786.
## 7 Afghanistan Asia      1982   39.9 12881816    978.
## 8 Afghanistan Asia      1987   40.8 13867957    852.
## 9 Afghanistan Asia      1992   41.7 16317921    649.
## 10 Afghanistan Asia      1997   41.8 22227415    635.
## # i 1,694 more rows
```

Elements of ggplot2

Hadley Wickham's ggplot2 is one of the most popular packages in the entire R canon.

- It also happens to be built upon some deep visualization theory: i.e. Leland Wilkinson's *The Grammar of Graphics*.

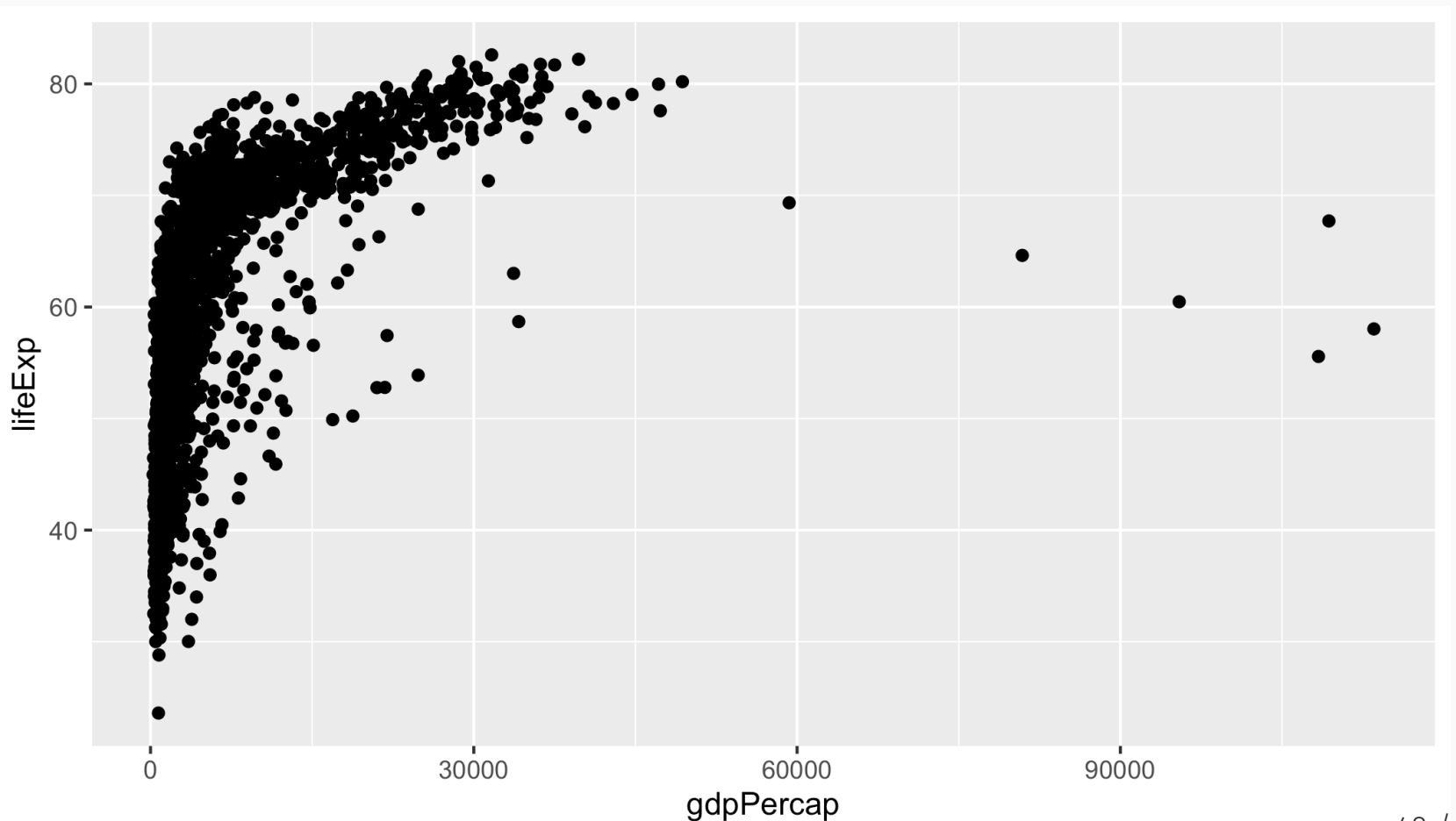
There's a lot to say about ggplot2's implementation of this "grammar of graphics" approach, but the three key elements are:

1. Your plot ("the visualization") is linked to your variables ("the data") through various **aesthetic mappings**.
2. Once the aesthetic mappings are defined, you can represent your data in different ways by choosing different **geoms** (i.e. "geometric objects" like points, lines or bars).
3. You build your plot in **layers**.

That's kind of abstract. Let's review each element in turn with some actual plots.

1. Aesthetic mappings

```
ggplot(data = gapminder, mapping = aes(x = gdpPercap, y = lifeExp)) +  
  geom_point()
```



1. Aesthetic mappings (cont.)

```
ggplot(data = gapminder, mapping = aes(x = gdpPercap, y = lifeExp)) +  
  geom_point()
```

Focus on the top line, which contains the initialising `ggplot()` function call. This function accepts various arguments, including:

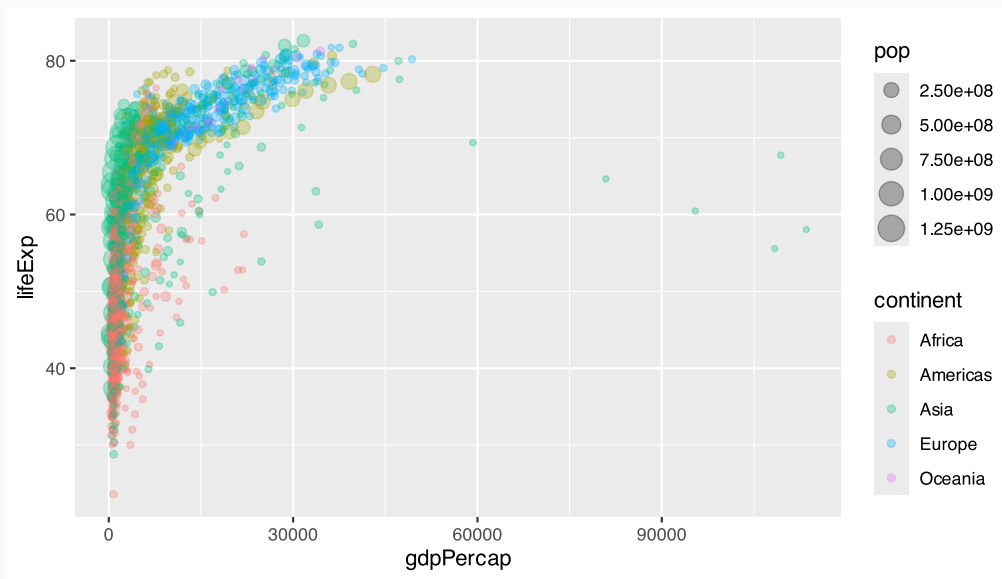
- Where the data come from (i.e. `data = gapminder`).
- What the aesthetic mappings are (i.e. `mapping = aes(x = gdpPercap, y = lifeExp)`).

The aesthetic mappings here are pretty simple: They just define an x-axis (GDP per capita) and a y-axis (life expectancy).

- To get a sense of the power and flexibility that comes with this approach, however, consider what happens if we add more aesthetics to the plot call...

1. Aesthetic mappings (cont.)

```
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp, size = pop, col = continent))\n  geom_point(alpha = 0.3) ## "alpha" controls transparency. Takes a value between 0 and 1
```

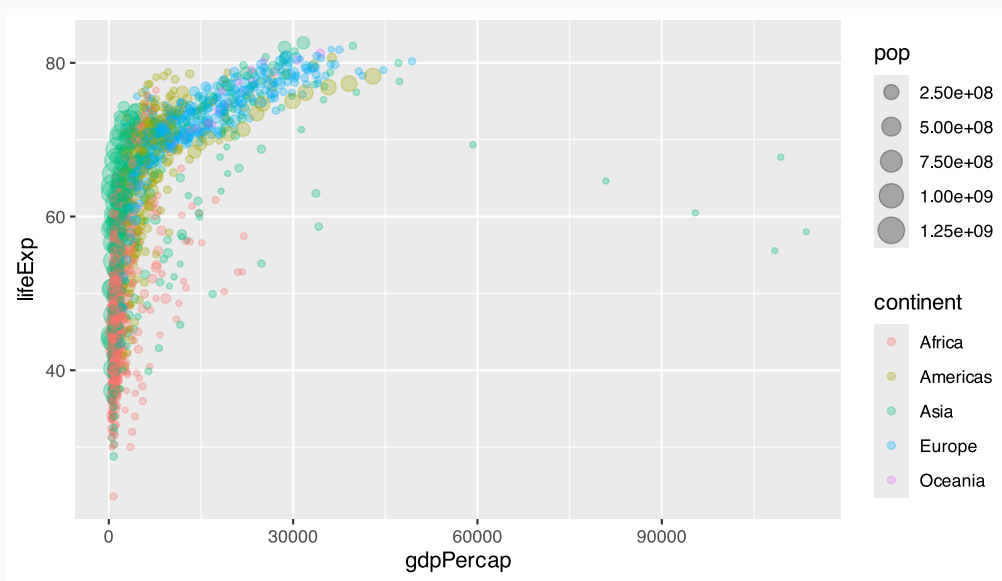


Note that I've dropped the "mapping =" part of the ggplot call. Most people just start with "aes(...)", since `ggplot2` knows the order of the arguments.

1. Aesthetic mappings (cont.)

We can specify aesthetic mappings in the geom layer too.

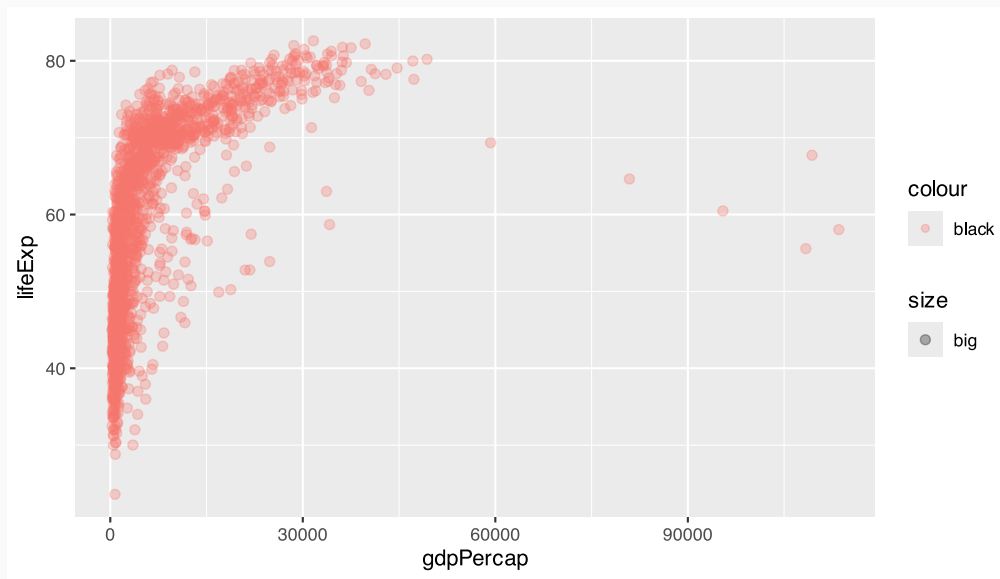
```
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp)) + ## Applicable to all geom:  
  geom_point(aes(size = pop, col = continent), alpha = 0.3) ## Applicable to this geom
```



1. Aesthetic mappings (cont.)

Oops. What went wrong here?

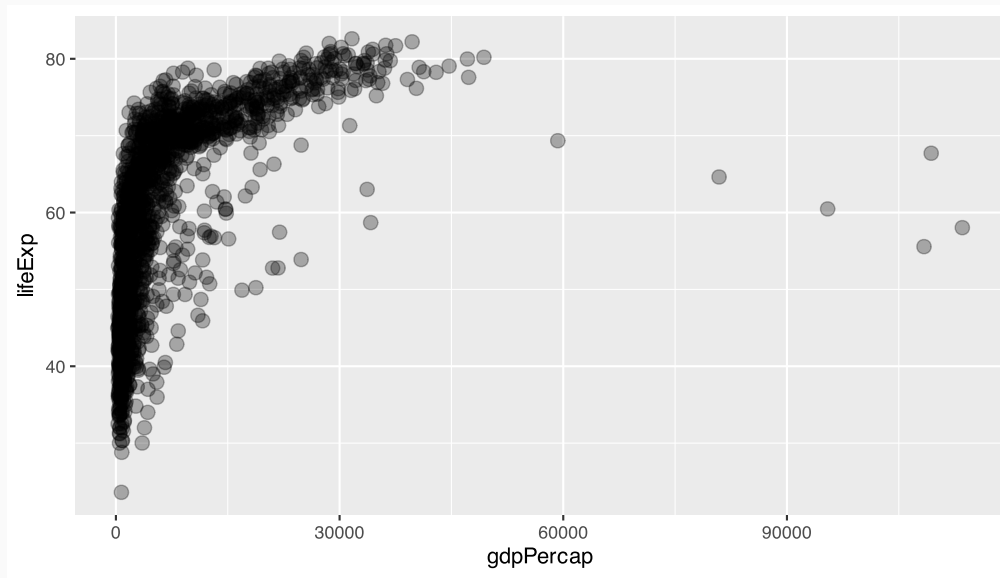
```
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp)) +  
  geom_point(aes(size = "big", col="black"), alpha = 0.3)
```



1. Aesthetic mappings (cont.)

This is what we wanted to do!

```
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp)) +  
  geom_point(size = 3, col="black", alpha = 0.3)
```

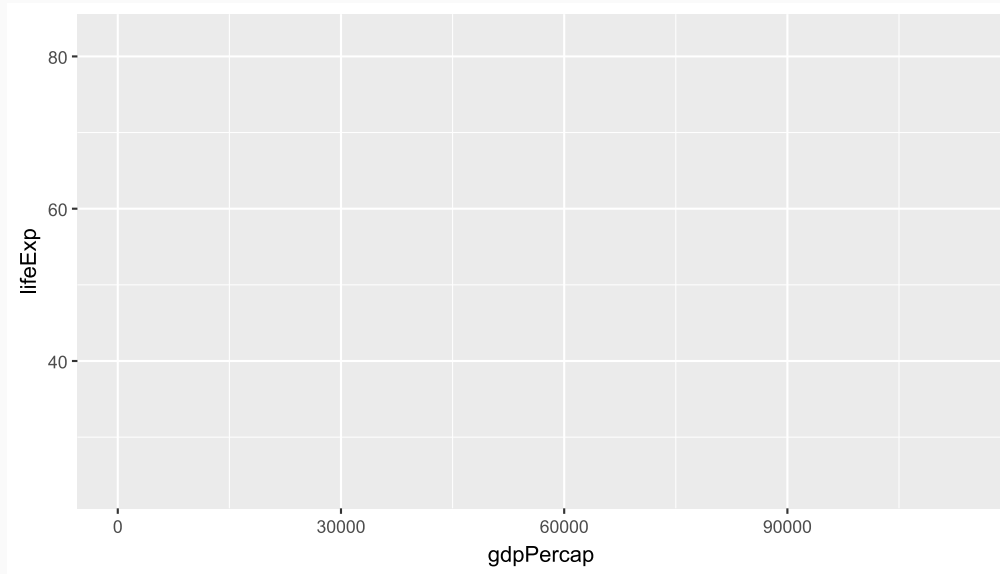


Note: `size` must take a numeric argument, not a character.

1. Aesthetic mappings (cont.)

At this point, instead of repeating the same ggplot2 call every time, it will prove convenient to define an intermediate plot object that we can re-use.

```
p = ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp))  
p
```

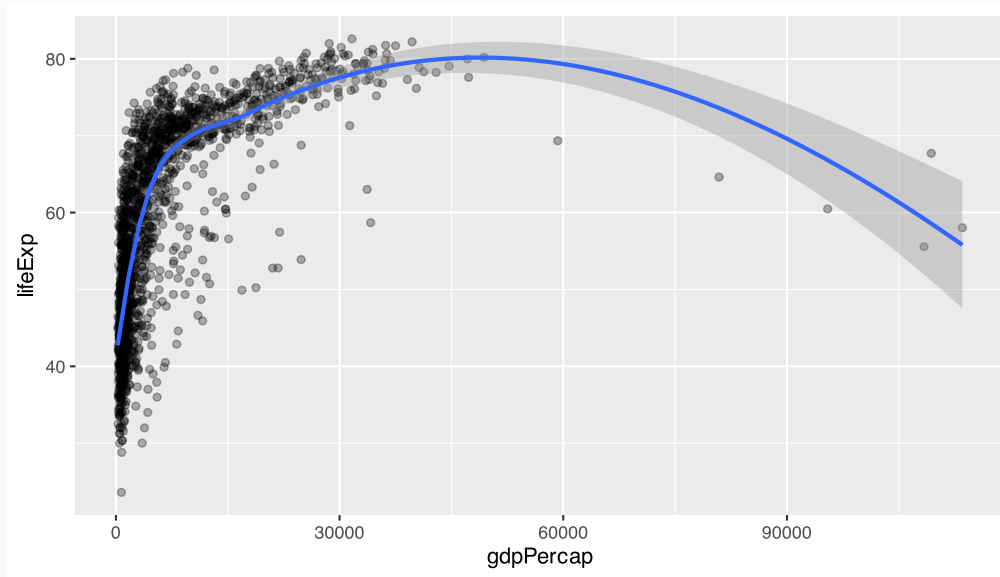


2. Geoms

Once your variable relationships have been defined by the aesthetic mappings, you can invoke and combine different geoms to generate different visualizations.

```
p + geom_point(alpha = 0.3) +  
  geom_smooth(method = "loess")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

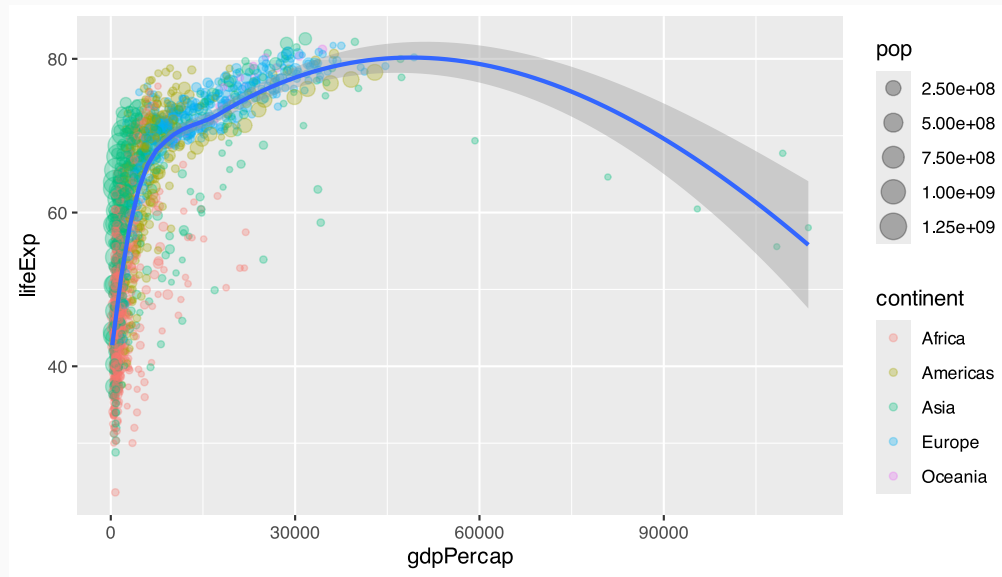


2. Geoms (cont.)

Aesthetics can be applied differentially across geoms.

```
p + geom_point(aes(size = pop, col = continent), alpha = 0.3) +  
  geom_smooth(method = "loess")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



2. Geoms (cont.)

The previous plot provides a good illustration of the power (or effect) that comes from assigning aesthetic mappings "globally" vs in the individual geom layers.

- Compare: What happens if you run the below code chunk?

```
p_bad = ggplot(data = gapminder,  
               aes(x = gdpPercap, y = lifeExp, size = pop, col = continent)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "loess")
```

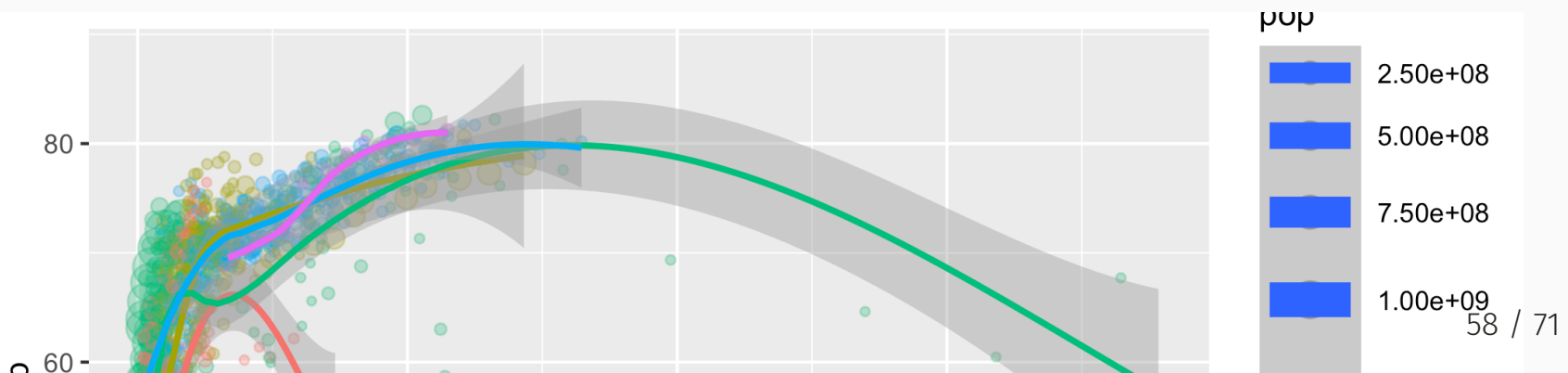
2. Geoms (cont.)

p_bad

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: size.  
## i This can happen when ggplot fails to infer the correct grouping structure in  
## the data.  
## i Did you forget to specify a `group` aesthetic or to convert a numerical  
## variable into a factor?
```



2. Geoms (cont.)

Similarly, note that some geoms only accept a subset of mappings. E.g. `geom_density()` doesn't know what to do with the "y" aesthetic mapping.

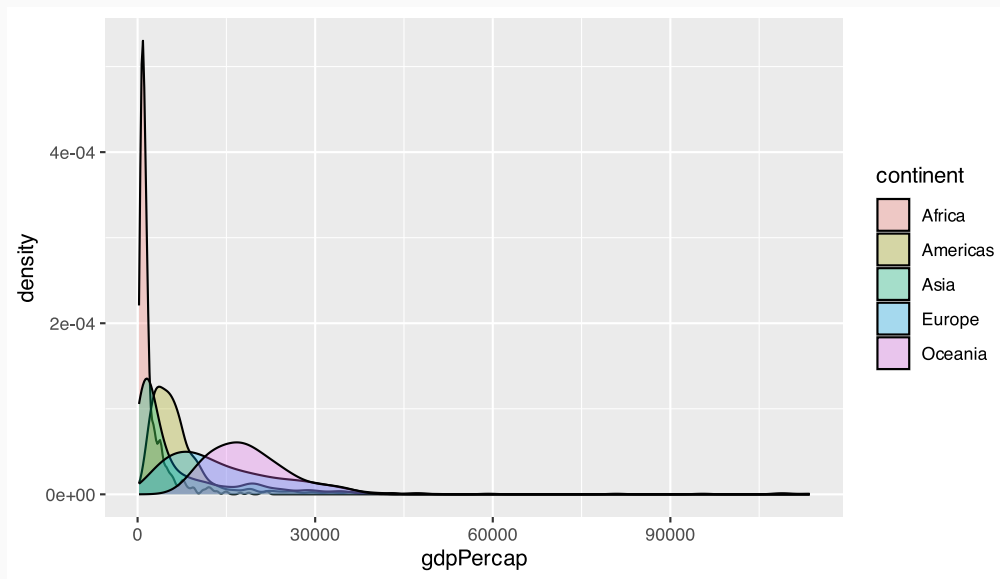
```
p + geom_density()
```

```
## Warning: The following aesthetics were dropped during statistical transformation: y.  
## ⓘ This can happen when ggplot fails to infer the correct grouping structure in  
##   the data.  
## ⓘ Did you forget to specify a `group` aesthetic or to convert a numerical  
##   variable into a factor?  
  
## Error in `geom_density()`:  
## ! Problem while setting up geom.  
## ⓘ Error occurred in the 1st layer.  
## Caused by error in `compute_geom_1()`:  
## ! `geom_density()` requires the following missing aesthetics: y.
```

2. Geoms (cont.)

We can fix that by being more careful about how we build the plot.

```
ggplot(data = gapminder) + ## i.e. No "global" aesthetic mappings"  
  geom_density(aes(x = gdpPercap, fill = continent), alpha=0.3)
```



3. Build your plot in layers

We've already seen how we can chain (or "layer") consecutive plot elements using the `+` connector.

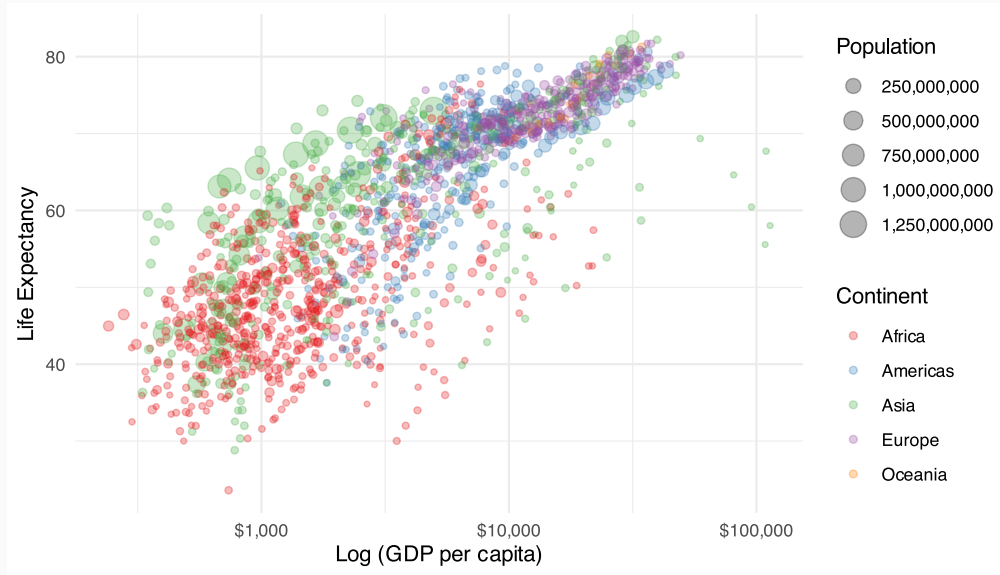
- The fact that we can create and then re-use an intermediate plot object (e.g. "p") is testament to this.

But it bears repeating: You can build out some truly impressive complexity and transformation of your visualization through this simple layering process.

- You don't have to transform your original data; ggplot2 takes care of all of that.
- For example (see next slide for figure).

```
p2 =  
  p +  
  geom_point(aes(size = pop, col = continent), alpha = 0.3) +  
  scale_color_brewer(name = "Continent", palette = "Set1") + ## Different colour scale  
  scale_size(name = "Population", labels = scales::comma) + ## Different point (i.e. size)  
  scale_x_log10(labels = scales::dollar) + ## Switch to logarithmic scale on x-axis. Use dollar  
  labs(x = "Log (GDP per capita)", y = "Life Expectancy") + ## Better axis titles  
  theme_minimal() ## Try a minimal (b&w) plot theme
```

3. Build your plot in layers (cont.)



What else?

We have barely scratched the surface of ggplot2's functionality... let alone talked about the entire ecosystem of packages that has been built around it.

- Here's are two quick additional examples to whet your appetite

Note that you will need to install and load some additional packages if you want to recreate the next two figures on your own machine. A quick way to do this:

```
if (!require("pacman")) install.packages("pacman")
```

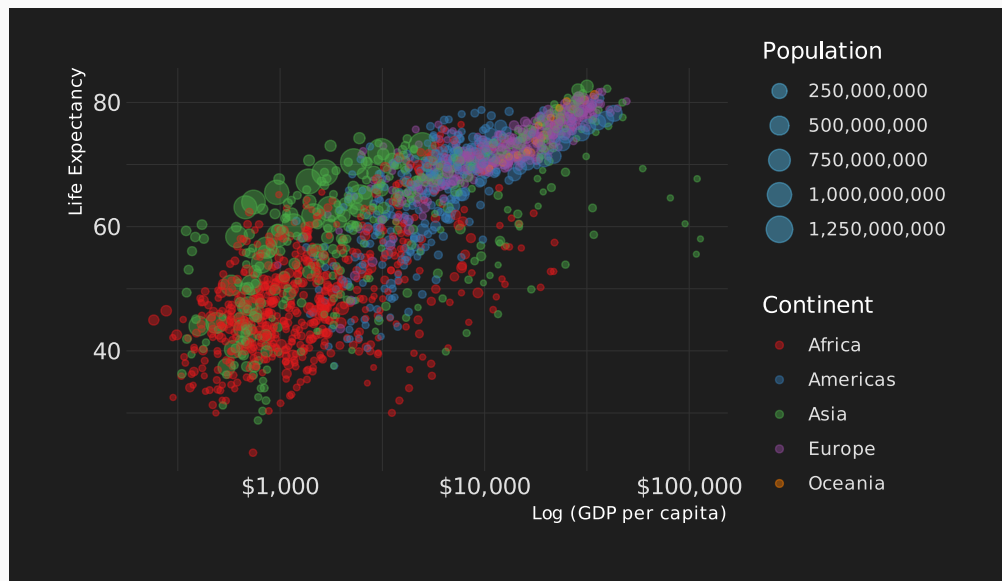
```
## Loading required package: pacman
```

```
pacman::p_load(hrbrthemes, gganimate)
```

What else? (cont.)

Simple extension: Use an external package theme.

```
# library(hrbrthemes)
p2 + theme_modern_rc() + geom_point(aes(size = pop, col = continent), alpha = 0.2)
```



What else? (cont.)

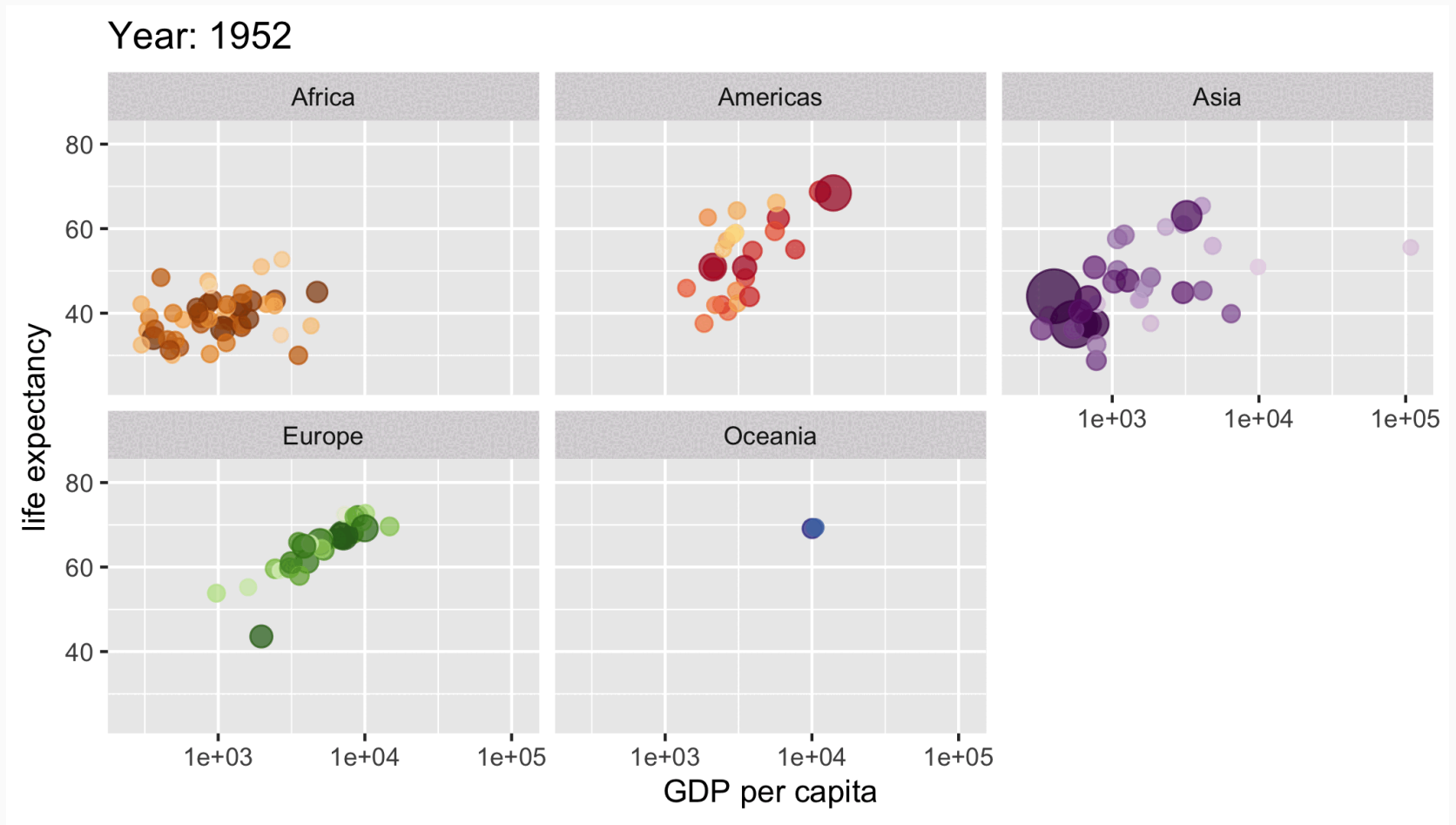
Elaborate extension: Animation! (See the next slide for the resulting GIF.)

```
library(gganimate)
library(magick)

ggplot(gapminder, aes(gdpPercap, lifeExp, size = pop, colour = country)) +
  geom_point(alpha = 0.7, show.legend = FALSE) +
  scale_colour_manual(values = country_colors) +
  scale_size(range = c(2, 12)) +
  scale_x_log10() +
  facet_wrap(~continent) +
  # Here comes the gganimate specific bits
  labs(title = 'Year: {frame_time}', x = 'GDP per capita', y = 'life expectancy') +
  transition_time(year) +
  ease_aes('linear')
```

What else? (cont.)

```
## Warning: package 'magick' was built under R version 4.3.3
```



Very cool. Also very useful! Note: GIF does not work with pdf format, use html instead.

What else? (cont.)

There's a lot more to say, but I think we'll stop now for today's lecture.

We also haven't touched on ggplot2's relationship to "tidy" data.

- It actually forms part of a suite of packages collectively known as the **tidyverse**.
- We will get back to this in later lectures.

Rest assured, you will be using ggplot2 throughout the rest of this course and developing your skills along the way.

In the meantime, I want you to do some reading and practice on your own. Pick either of the following (or choose among the litany of online resources) and work through their examples:

- **Chapter 3** of *R for Data Science* by Hadley Wickham and Garrett Grolemund.
- *Data Visualization: A Practical Guide* by Kieran Healy.
- *Designing ggplots* by Malcom Barrett.

Next lecture: Getting started with R.

Appendix

R vs Stata

If you're coming from Stata, some additional things worth emphasizing:

- Multiple objects (e.g. data frames) can exist happily in the same workspace.
 - No more `keep`, `preserve`, `restore` hackery. (Though, props to [Stata 16](#).)
 - This is a direct consequence of the OOP approach.
- You will load packages at the start of every new R session. Make peace with this.
 - "Base" R comes with tons of useful in-built functions. It also provides all the tools necessary for you to write your own functions.
 - However, many of R's best data science functions and tools come from external packages written by other users.
- R easily and infinitely parallelizes. For free.
 - Compare the cost of a [Stata/MP](#) license, nevermind the fact that you effectively pay per core...
- You don't need to `tset` or `xtset` your data. (Although you can too.)

Base R plot

```
par(mar = c(4, 4, 1, .1)) ## Just for nice plot margins on this slide deck  
plot(mtcars$wt, mtcars$mpg)  
abline(fit, col = "red")
```

